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respectively suffer one dimensional elastic collision. Find their velocities after collision.

**Solution:**

Mass $m_1 = 60$ kg

Mass $m_2 = 30$ kg

$$V_1 = 40\ cms^{-1}$$

$$V_2 = 30\ cms^{-1}$$

**Solution:**

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2$$

$$v_2 = \frac{(m_2 - m_1)}{m_1 + m_2}u_2 + \frac{2m_1}{m_1 + m_2}u_1$$

Substituting the values, we get,

$$v_1 = \frac{(60-30)}{90} \times 40 + \frac{2 \times 30}{90} \times 30$$

$$v_1 = \frac{1}{90}\ [1200 + 1800]$$

$$= \frac{3000}{90} = 33.3 cms^{-1}$$

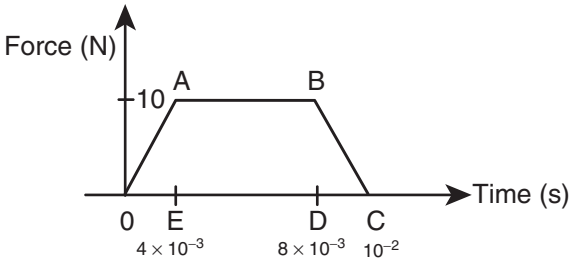
Likewise,

$$v_2 = \frac{(30-60)}{90} \times 30 + \frac{2 \times 60}{90} \times 40$$

$$v_2 = \frac{1}{90}[-900 + 4800]$$

$$= \frac{3900}{90} = 43.3 cms^{-1}$$

6. A particle of mass 70 g moving at 50 cms$^{-1}$ is acted upon by a variable force as shown in the figure. What will be its speed once the force stops?




**Solution:**

The area under the graph gives the impulse.

Impulse I = area of ΔOAE+ area of rectangle ABDE+ area of ΔDBC

$$I = \frac{1}{2} \times 4 \times 10^{-3} \times 10 + (8-4) \times 10^{-3} \times 10 + \frac{1}{2} \times 10 \times 10^{-2}(1 - 0.8)$$

$$2 \times 10^{-2} + 4 \times 10^{-2} + 1 \times 10^{-2}$$

$$= 7 \times 10^{-2}\ kgms^{-1}$$

But Impulse = 2 × initial momentum of the particle= 2 × m × u

$$2mu = 7x10^{-2}$$

$$u = \frac{7 \times 10^{-2}}{2 \times m} = \frac{7 \times 10^{-2}}{2 \times 70 \times 10^{-3}} = 50 cms^{-1}$$

Hence the particle will reverse its direction and move with its initial speed.

7. A particle strikes a horizontal frictionless floor with a speed u at an angle θ with the vertical and rebounds with the speed v at an angle $\phi$ with an vertical. The coefficient of restitution between the particle and floor is e. What is the magnitude of v?

